

CS 401/501 Homework 3 - Spring 2026

Pre-Reqs

1. Install GCC for your system
2. Install MPICC for your system
3. If you are on Windows it is recommended you use WSL

Tasks

Task 1

The only task is to complete parallel

filtering.c . The goal is to blur an image given various stencil sizes in using parallel computation.

Task 2

Write a report talking about:

Program Design: Reflections on your approach

Parallel I/O: The difference between reading/writing files in parallel and serially

Ghost Cell Exchange: What is it?, did you use it in your approach and if so, how?

Challenges, Lessons Learned and Future Improvements: Reflections on the homework

Screenshots: Screenshots of your program time

Hints

To visualize the .raw image you can use either <https://uabacmrawimage.web.app> or <https://www.paraview.org/>. The dimensions are 2560 x 2560

The bigger the stencil size the more blurry your image should be (we will test your code using various stencil sizes)

We have given you a serial

filtering.c that you can do to test potential blurring algorithms without parallelization, however, this file is not something you have to complete nor turn in

Running the code using MPI

```
# Compile the file using mpicc  
mpicc -o parallel
```

```
filtering parallel
```

```
filtering.c  
# Run the executable  
mpirun -np 4 ./parallel
```

```
filtering -g 2560x2560x1 -s 2 -f data/brain
```

2560

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Checkout the parse

—

args for an explanation of what each arg does

2560.raw

SubmissionBLAZERID_CS401_HW3.zip

data/

YOUR BLURRED FILES (.png or .raw)

Please name it: brain_2560_2560_blurred.raw

YOUR INPUT FILE (brain_2560_2560.raw)

CITATIONS.md (with your work citations)

BLAZERID_report.pdf (notice the pdf)

parallel_filtering.c

ICF (.pdf or .docx)

[OPTIONAL] serial_fitering.c

Rubric

Category	Criterion	Weight (%)	Description
Report	Clarity and Organization	5	Well-structured report with clear sections (e.g., Name, Introduction, Methodology, Results, Discussion).
Report	Design & Implementation	5	Detailed explanation of program design, parallel I/O, ghost cell exchange, and blurring algorithm.
Report	Performance Analysis	5	Timing results for I/O, communication, and computation phases with insights on efficiency and scaling.
Report	Discussion & Reflection	5	Reflection on challenges, lessons learned, and potential future improvements.
Correctness & Functionality	Parallel I/O	10	Correct computation of file offsets and parallel reading/writing of image segments using MPI/POSIX I/O.
Correctness & Functionality	Ghost Cell (Halo) Communication	10	Proper exchange of ghost cells between processes, addressing boundary cases effectively.
Correctness & Functionality	Filtering (Blurring) Operation	10	Correct application of the blurring algorithm using neighbor data, including ghost cells as needed.
Correctness & Functionality	Error Handling & Robustness	10	Validation of command-line arguments and graceful error handling throughout the code.

Performance & Scalability	Timing and Profiling	10	Accurate measurement of execution time (using MPI_Wtime) for different phases of the program.
Performance & Scalability	Scalability Analysis	10	Demonstration and discussion on how the program scales with an increasing number of processes.
Code Quality & Documentation	Code Organization & Readability	10	Clear, well-organized, and modular code with appropriate comments and documentation.
Code Quality & Documentation	Adherence to Assignment Requirements	10	Proper use of MPI functions and compliance with assignment specifications and README guidelines.